Global Temperature Trends: Random Walk or Deterministic Warming?

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Abstract

This project analyzes global temperature anomalies (1880–present) to distinguish between random walk behavior and a deterministic warming trend. Using seasonal differencing and unit root tests, I find that the series is stationary after removing seasonal effects, providing evidence against a random walk model. The remaining dependence is well captured by a low-order ARMA(1,1) model applied to the seasonally differenced data.

GitHub Repository: *https://github.com/XI-YW/STAT354.git*

1 Introduction

Understanding the dynamics of global temperature is a central problem in climate science. A key statistical question is whether global temperature anomalies follow a *random walk* or exhibit a *deterministic warming trend*. In a random walk, shocks have permanent effects, whereas in a deterministic trend model, deviations from the trend are temporary and mean-reverting.

In this project, I analyze the NASA GISS Surface Temperature Analysis dataset, which provides monthly global temperature anomalies from 1880 to the present. My objective is to determine whether the observed warming pattern is better described as a random walk or as a deterministic trend with seasonal structure.

2 Data Description

The dataset used for this project is the NASA GISS Surface Temperature Analysis (GISTEMP), which reports global temperature anomalies relative to a baseline period. I use monthly observations from 1880 to the present.

Let x_t denote the monthly temperature anomaly at time t. The series exhibits a clear upward trend, strong seasonal variations, and irregular fluctuations.

3 Exploratory Analysis

Figure 1 shows the time series plot of the monthly temperature anomalies. The series exhibits a clear upward trend along with regular seasonal variations.

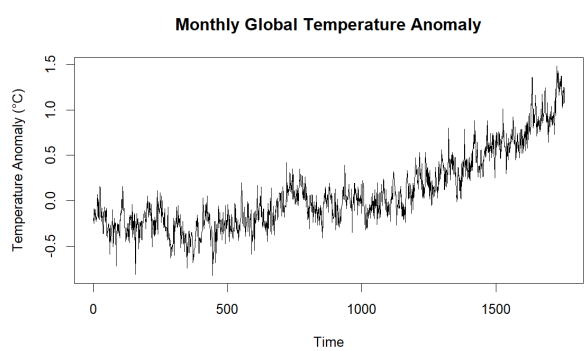

Figure 1: Monthly global temperature anomalies.

The autocorrelation function (ACF) of the original series (Figure 2) shows strong persistence and slow decay, indicating nonstationarity.

To remove recurring annual effects, I apply seasonal differencing with period $s = 12$:

$$(1 - B^{12})x_t.$$

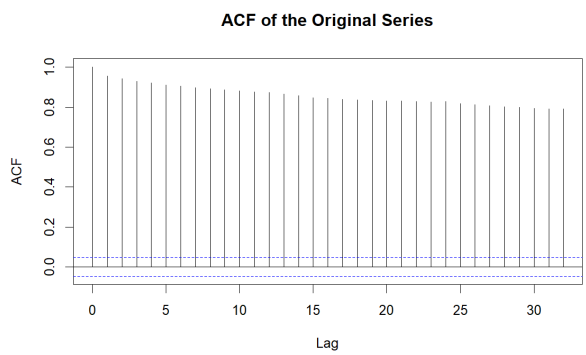

Figure 2: ACF of the original series.

This transformation compares each month's temperature anomaly with the anomaly from the same month in the previous year. If the series contains a stable yearly seasonal pattern, seasonal differencing removes much of that repeated structure. In this setting, using $s = 12$ is natural because the data are monthly and the annual cycle corresponds to 12 observations.

After seasonal differencing, the transformed series is more stable in level, and its ACF (Figure 3) shows a much weaker seasonal pattern. This suggests that a substantial part of the apparent nonstationarity in the original data is due to seasonality.

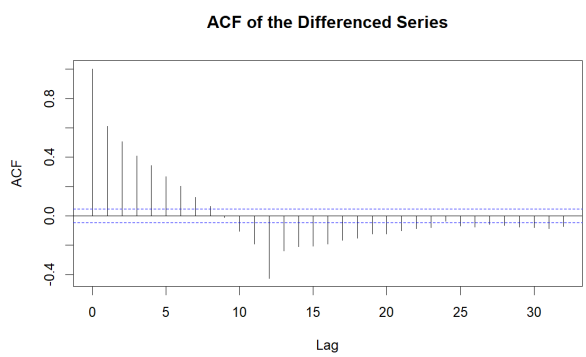

Figure 3: ACF after seasonal differencing.

However, the ACF of the differenced series still shows several autocorrelations outside the confidence bounds at low lags. This indicates that the transformed series is not white noise, but instead retains short-run dependence. This behavior is expected and suggests that the remaining dependence structure can be captured using a low-order ARMA model.

4 Unit Root Testing

I apply the Augmented Dickey–Fuller (ADF) test and the Phillips–Perron test to the seasonally differenced series. Both tests strongly reject the null hypothesis of a unit root ($p < 0.01$), indicating that the series is stationary after removing seasonal effects.

This result suggests that the apparent nonstationarity in the original series is largely driven by deterministic components, including trend and seasonality, which provides evidence against modeling the temperature anomaly series as a pure random walk process.

5 Model Fitting

5.1 ARMA Model on the Differenced Data

To model the differenced series, I first performed a grid search over ARMA(p, q) models by looping over a range of values for p and q and selecting the model with the lowest AIC. This approach frequently selected models with relatively large orders, such as $(p, q) = (10, 10)$, which resulted in very low AIC values.

However, these high-order models are likely to be overfitted. They introduce many parameters and may capture noise rather than the underlying structure of the data, making them difficult to interpret and less reliable for inference.

Moreover, the results from the grid search are inconsistent with the earlier exploratory analysis. The ACF of the seasonally differenced series shows a gradual decay pattern typical of a relatively low-order autoregressive process, suggesting that the appropriate model should have relatively small values of p and q. This inconsistency indicates that relying solely on AIC can lead to unnecessarily complex models.

To address this issue, I instead used an automatic model selection procedure that balances goodness-of-fit with model complexity. Based on the result, I selected an ARMA(1,1) model, so

the original time series can be equivalently represented by an ARIMA model applied after seasonal differencing:

$$(1 - B^{12})x_t = \phi_1(1 - B^{12})x_{t-1} + w_t + \theta_1 w_{t-1}.$$

This model provides a simpler and more interpretable representation of the data, capturing the short-run dependence that remains after seasonal differencing while avoiding overfitting. The selected model is also consistent with the gradual decay observed in the autocorrelation function.

5.2 ARIMA Model on the Original Data

For comparison, I also fit an ARIMA model to the original series using automatic model selection. Although seasonal terms were allowed, the selected model did not include any seasonal components. The selected model is ARIMA(3,1,3) with drift. This suggests that explicit seasonal structure is not substantial in the temperature anomaly series. The inclusion of a differencing term ($d = 1$) and a drift parameter reflects strong persistence in the series and allows the model to capture long-run upward movement in the data.

6 Model Comparison and Diagnostics

The residual diagnostics for both models are shown in Figures 4 and 5.

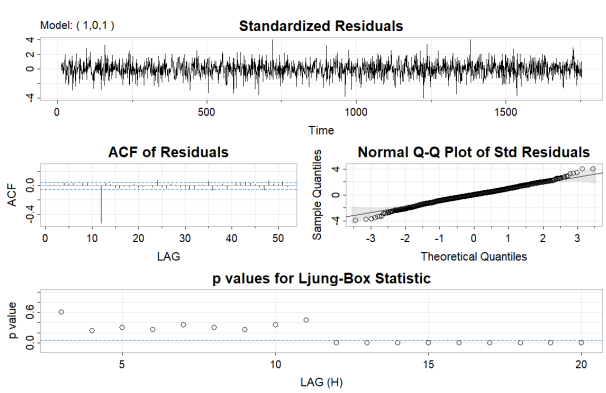

Figure 4: Residual diagnostics for ARMA(1,1) model.

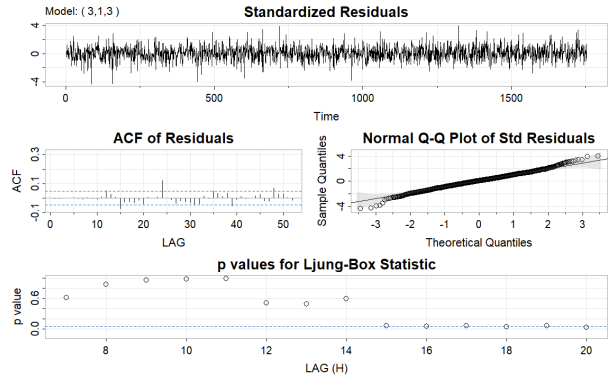

Figure 5: Residual diagnostics for ARIMA(3,1,3) model.

The ARMA(1,1) model captures most of the dependence structure, although the residual diagnostics indicate slight remaining seasonal correlation at lag 12. In contrast, the ARIMA(3,1,3) model provides improved residual behavior, with residuals that more closely resemble white noise.

Although the ARIMA(3,1,3) model includes a first-difference term ($d = 1$), this does not contradict the unit root test results. Rather, the use of differencing reflects the strong persistence and trend in the series, which can lead model selection procedures to favor differencing even in the absence of a true unit root.

The two models emphasize different aspects of the data. The ARMA(1,1) model demonstrates that, after removing seasonal effects, the series behaves as a stationary process with short-run dependence. In contrast, the ARIMA(3,1,3) model provides a better fit to the original data by allowing additional flexibility and capturing stronger persistence.

Although the ARIMA(3,1,3) model provides improved residual diagnostics, I prefer the ARMA(1,1) model applied to the seasonally differenced series. This model is more interpretable, aligns with the smooth decay observed in the autocorrelation function, and is consistent with the unit root test results. In contrast, the ARIMA model introduces additional complexity and may capture noise rather than the underlying structure of the data.

7 Conclusion

In this project, I investigated whether global temperature anomalies follow a random walk or exhibit a deterministic warming trend.

The results show that the series becomes stationary after removing seasonal effects and is well described by a low-order ARMA model. Although the ARIMA(3,1,3) model captures strong persistence and provides a slightly improved fit, the evidence from unit root testing and autocorrelation analysis indicates that the series does not follow a pure random walk.

Overall, I conclude that the temperature series is best characterized by a deterministic warming trend combined with seasonal variation and persistent but mean-reverting dynamics. Among the models considered, the ARMA(1,1) model applied to the seasonally differenced series provides the most appropriate balance between simplicity, interpretability, and consistency with the observed autocorrelation structure.

8 References

References

[1] GISTEMP Team (2026): *GISS Surface Temperature Analysis (GISTEMP), version 4*. NASA Goddard Institute for Space Studies. https://data.giss.nasa.gov/gistemp/data_v4.html.

[2] Shumway, R. H., & Stoffer, D. S. (2017). *Time Series Analysis and Its Applications: With R Examples, Fourth Edition*. Springer.

9 GAI Use Disclosure

- **What I used GAI for:** (1) Debugging; (2) requesting help for model selection; (3) asking about how to upload images to the LaTeX file; (4) asking about how to adjust the size of images and the font size of captions; (5) asking about the referencing format; (6) solving GitHub issues,
- **The specific tool(s) I used:** ChatGPT
- **What prompts I used to get the results:** (1) "<error message> how to solve this"; (2) "<code> should I make the range wider if I got p, q = 10" (3) "How to upload images to the overleaf"; (4) "How to change the size of the images in latex, the images seem to be small"/"How to make the size of captions smaller"; (5) "How to write the reference for this data https://data.giss.nasa.gov/gistemp/data_v4.html"; (6) "how to add a folder to my github main".